



Occurrence, Multidrug Resistance, and Multilocus Sequence Typing of Extended-Spectrum Beta-Lactamase/AmpC-Producing *Escherichia coli* from Farmed Eggs

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Abstract

The present study evaluated the occurrence, antibiogram profile, and sequence types (STs) of multidrug resistant (MDR) *Escherichia coli* from freshly laid eggs ($n = 480$), feed ($n = 24$), water ($n = 24$), poultry droppings ($n = 24$), and hand swab samples ($n = 10$) collected from 24 deep litter (DL) and caged poultry layer farms (12 per category) across Punjab, India. The overall *E. coli* contamination rate in DL and cage farms was 32% (95% confidence intervals [CI], 26.6–37.8%) and 16.7% (95% CI, 12.6–21.6%), respectively. The logistic regression analysis revealed that the DL system had higher odds of occurrence (odds ratio [OR]) of extended-spectrum beta-lactamase (ESBL) (2.195, 95% CI, 1.065, 4.522) and ESBL/AmpC coproducers (2.69, 95% CI, 1.122, 6.45) compared to the cage system. Additionally, isolates from the DL were 4.065 (95% CI, 1.477, 11.188) times more tetracycline resistant compared to the latter; however, resistance to amoxycyclavulanate (OR, 0.437; 95% CI, 0.209, 0.912), and ampicillin (OR, 0.343; 95% CI, 0.163, 0.720) was lesser in DL system. Notably, around 97.7% and 87.2% of the isolates from the DL and cage system were MDR, with the DL system having 6.439 (95% CI, 1.246, 33.283) times more chances of harboring MDR *E. coli*. Additionally, among the resistance genes, the DL system demonstrated significantly high presence of *bla*_{AmpC} (56%), *qnrA/B/S* (42.3%), and *tetA/B* (30.6%). Furthermore, multi-locus sequence typing of 11 MDR isolates ($n = 5$, DL, and 6, cage) revealed the presence of 10 STs, of which ST10, ST155, and ST156 were found to be of public health importance. Therefore, the present study highlights the burden of MDR, ESBL, and AmpC-producing *E. coli* on poultry eggs and farm environment, which could be carried over to human handlers and consumers upon direct contact during handling and processing.

Keywords: extended-spectrum beta-lactamase (ESBL) *E. coli*, AmpC *E. coli*, egg, deep litter, cage system

Introduction

Antimicrobial resistance has emerged as a global public health concern and the silent pandemic of the 21st century (Crowther-Gibson et al., 2011; Founou et al., 2021). The contributing factors include the injudicious use of antibiotics in the clinical treatment of humans and animals and applications in

food production (Llor and Bjerrum, 2014). The accretion of resistance against different classes of antibiotics has resulted in the emergence of multidrug-resistant (MDR) phenotypes, of which resistance against priority pathogens is of prime importance.

The World Health Organization (WHO) categorized members of *Enterobacteriaceae* especially extended-spectrum

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beta-lactamase (ESBL) producing *Escherichia coli* as a critical pathogen (WHO, 2024), owing to the associated clinical implications and urgent need for new antimicrobials. Due to their clinical significance in Indian healthcare settings, ESBL and AmpC-producing *E. coli* achieved the status of high-priority pathogens (DBT, 2021). While these bacteria were earlier associated with human hospital settings and institutional care centers, their increasing incidence in food-producing animals and retail meat suggests a possible gene exchange between different reservoirs (Wu et al., 2018).

India is the second largest poultry egg producer worldwide with 4.63% of production from Punjab (FAOSTAT 2021, <http://www.fao.org/faostat/en/#data/QC>) with the state of Punjab contributing to 4.63% of the nation's total egg (FAOSTAT, 2021). The layer farms in Punjab practice free range/deep litter (DL) and commercial cage farming. The management practices under these systems differ with respect to exposure with the outer surroundings, veterinary supervision, farm biosecurity (BS), and hygienic practices at farm. While multiple studies have focused on ESBL, AmpC *E. coli* from poultry cloacal swabs (Brower et al., 2017; Tewari et al., 2019), feces (Mondal et al., 2023; Senthamilselvan et al., 2024; Sharma et al., 2021), and caeca (Sakthikarthikeyan et al., 2024), similar studies on eggs and farm environment are scarce in number. Further, the absence of continuous surveillance of eggs for ESBL/AmpC *E. coli*, in addition to noncompliance with hygiene and BS measures from farm to fork augments the risk of cross-contamination from eggs to clean food items, surfaces, and human handlers with these resistant strains. Therefore, the present study addresses this critical gap and investigates farmed eggs, environmental, and hand swab samples for contamination with ESBL, AmpC *E. coli*, MDR, molecular characterization, and multilocus sequence typing of resistant isolates in DL and caged poultry layer farms across Punjab, India.

Materials and Methods

Sample collection

Twenty-four poultry layer farms distributed equally between DL and cage system (Cg) from Punjab were randomly selected using convenient and purposive sampling. Twenty freshly laid eggs from each farm ($n = 480$) were aseptically collected in disposable sampling bags (HiMedia Lab, Mumbai). Around five different feed samples from different locations inside the farm were collected and pooled to form one feed sample representative of one farm. Similarly, water and poultry dropping samples were collected; a total of 72 environmental samples were collected. The hand swab samples from farm handlers were collected after informed consent; however, samples were limited ($n = 10$) due to the unwillingness of farmers from all the farms. All the samples were brought to the laboratory on ice coolers maintained at 4°C and either processed the same day or kept at 4°C (eggs) or −20°C (environmental samples) for further analysis.

Post sample collection, the farmers were enquired about the management practices, usage of antibiotics, hygiene, and BS measures using a well-structured and informed questionnaire (Supplemental Data S1).

Isolation and identification of *E. coli*

Each egg was cracked open on the air-sac end using a sterile glass rod to separate and empty the inner content in an aseptic disposable sampling bag. The remaining eggshell was

put in another sterile disposable bag containing 20 mL of buffered peptone water and hand-crushed. Around 5 mL each of eggshell wash and inner content were enriched separately in 45 mL brain–heart infusion broth (BHI); however, water (5 mL), feed (12.5 g), and droppings (12.5 g) were enriched in 45 mL, 112.5 mL, and 112.5 mL of BHI broth, respectively. The swab tips along with their diluent were enriched in 9 mL of BHI broth. All the samples were incubated overnight at 37°C.

Following enrichment, a loopful of samples was plated on eosin methylene blue agar and incubated at 37°C for 24–48 h. The round colonies with lustrous metallic sheen were picked, and a single colony was inoculated on BHI agar for purification. The purified colonies were subjected to Gram's staining, followed by matrix-assisted laser desorption/ionization-time of flight mass spectrophotometry (Bruker Daltonik, Germany) for confirmation of *E. coli*.

Antimicrobial sensitivity testing of *E. coli* isolates

The antibiotic sensitivity of *E. coli* isolates was performed using the Kirby-Bauer disc diffusion method, following the Clinical and Laboratory Standards Institute (CLSI, 2020) guidelines. The sensitivity of isolates was tested against 15 antibiotics; ceftazidime (CAZ, 30 µg), ciprofloxacin (5 µg), levofloxacin (5 µg), amikacin (30 µg), gentamicin (10 µg), tobramycin (10 µg), amoxycylav (20/10 µg), ampicillin (10 µg), meropenem (10 µg), trimethoprim (5 µg), cefazolin (30 µg), cotrimaxazole (25 µg), chloramphenicol (30 µg), and tetracycline (30 µg). *E. coli* strain ATCC 25923 was used as a quality control strain. The isolates were marked as sensitive, intermediate, and resistant per the CLSI (2020) guidelines; however, those depicting intermediate resistance were considered resistant for the analyses. The isolates exhibiting resistance against >3 antimicrobials belonging to at least two classes were categorized as MDR. Furthermore, the multiple antibiotic resistance index (MAR) was calculated using the given formula (Krumperman 1983).

$$\text{MAR} = \frac{\text{number of antibiotics to which the isolate was resistant (a)}}{\text{number of antibiotics to which the isolate was exposed (b)}}$$

Here, $b = 15$.

Detection of ESBL production (modified disc diffusion test)

A sterile MHA plate was swabbed with test culture and dried. Subsequently, cefotaxime (CTX 30 µg), CAZ (30 µg), CTX with clavulanic acid (CEC 30/10 µg), and CAZ with clavulanic acid (CAC 30/10 µg) discs were placed at a distance of 24 mm from each other. All the discs were inoculated with phenylboronic acid (PBA) (400 µg) and 0.1M EDTA (292 µg) and incubated at 37°C for 18–24 h. Following incubation, the difference in the zone of inhibition (ZOI) around CTX, CEC and CAZ, CAC was calculated; a difference ≥5 mm indicated ESBL producer (Poulou et al., 2014).

Detection of AmpC production

A sterile MHA plate was swabbed with cefoxitin-sensitive *E. coli* ATCC 25923 strain and dried. Subsequently, a cefoxitin disc (FOX) was gently placed on an MHA plate using sterile forceps. Simultaneously, sterile filter discs impregnated with tris-EDTA (AmpC disc) were inoculated with the test culture and placed touching the cefoxitin disc. The plate was incubated

at 37°C for 18–24 h. Following incubation, the plate was observed for any indentation/flattening of ZOI around FOX near the inoculated AmpC disc; flattening/indentation indicated AmpC production (Poulou et al., 2014).

Detection of Klebsiella pneumoniae carbapenemase production

A sterile MHA plate was swabbed with the test culture, followed by placement of a plain meropenem disc (MRP 10 µg) and meropenem disc (10 µg) with phenylboronic acid (PBA) (400 µg), and incubation at 37°C for 18–24 h. Following incubation, the difference between the ZOI of MRP and MRP-PBA combination was calculated; a difference ≥5 mm indicated *Klebsiella pneumoniae* carbapenemase (KPC) production (Poulou et al., 2014).

Detection of metallo-beta-lactamase production

The test culture was swabbed over a sterile MHA plate, followed by the placement of two MRP discs (10 µg). 0.5M EDTA was added to one of the discs, and the plate was incubated overnight at 37°C. The difference in ZOI between the MRP-EDTA combination and MRP alone was calculated; a difference ≥5 mm indicated metallo-beta-lactamase (MBL) producer (Poulou et al., 2014).

Molecular characterization of antibiotic resistance genes in E. coli isolates

The isolation of genomic DNA from confirmed isolates was performed using the Qiagen DNA extraction kit using the manufacturer's protocol. The single-plex PCR reaction for all antibiotic resistance genes in the test *E. coli* isolates was performed in Mastercycler Gradient Thermocycler (Eppendorf, Germany) using a predefined set of primers and cycling conditions (Supplemental Data S1). Further DNA templates from the previously identified isolates were used as positive controls: *bla*_{TEM} (GenBank:OL629040.1), *bla*_{SHV} (GenBank:OL587692.1), and *bla*_{AmpC} (GenBank:OL587697.1). Furthermore, the field isolates from the present study were used as positive controls for other antibiotic resistance genes.

Hierarchical clustering of AmpC-producing E. coli

The AmpC-producing *E. coli* were clustered on the basis of their phenotypic resistance against penicillins, cephalosporins, and sulfamethoxazole-trimethoprim combination using Ward's linkage and squared Euclidean distance measure.

Multilocus sequence typing of MDR E. coli isolates

A total of 11 MDR, ESBL-producing *E. coli* isolates from eggs and poultry droppings from DL and cage farms were randomly selected and processed for whole genome sequencing (WGS). Thereby sequences of seven housekeeping genes (HKGs); *adh*, *fumC*, *icd*, *purA*, *gyrB*, *recA*, and *mdh* were blasted in PubMLST database to deduce the allelic profile and sequence types (STs). The data of whole genome were submitted to GenBank database and accession numbers were granted.

Statistical analysis

The statistical analysis was conducted using IBM SPSS Statistics v.27.0.1 (IBM Corporation, Armonk, NY). The occurrence

of *E. coli* from different sample sources, phenotypic and genotypic AMR profile, and MDR% between different systems of rearing was compared using chi-square statistic, where $p < 0.05$ was taken as significant. Further, the modeling of resistance profiles of isolates; phenotypic/genotypic/MDR (dependent variable) versus system of rearing (independent variable) was done using binary logistic regression analysis and odds ratio (OR) was calculated and plotted using forest plot.

Results

Occurrence of E. coli

The sample-wise *E. coli* contamination from different systems of rearing is tabulated in Table 1. The analysis revealed that the *E. coli* contamination was significantly affected by the system of poultry rearing ($p < 0.05$), with the eggs from DL being 1.97 (95% CI, 1.289, 3.009) times more contaminated than those in Cg. Additionally, the overall occurrence of *E. coli* from all samples in DL was higher (OR, 2.125; 95% CI, 1.420, 3.181) than Cg.

Occurrence of ESBL, AmpC, and ESBL/AmpC coproducing E. coli

DL harbored significantly more ESBL and ESBL/AmpC coproducing *E. coli* ($p < 0.05$) (Table 1), while no difference in AmpC *E. coli* was noted between the systems ($p > 0.05$). The eggs in DL harbored more ESBL *E. coli* (OR, 2.429; 95% CI, 1.135, 5.196) and coproducers (OR, 3.449; 95% CI, 1.1291, 9.211) compared to those in Cg. Furthermore, the overall occurrence of ESBL *E. coli* and ESBL/AmpC coproducers was 2.195 (95% CI, 1.065, 4.522) and 2.69 (95% CI, 1.122, 6.45) times more in the DL system than Cg (Fig. 1).

Phenotypic antibiogram profile of E. coli isolates

The isolates from eggs in DL exhibited maximum resistance against tetracycline (92%), cefazolin (76%), and CAZ (68%). Likewise, in Cg, egg isolates were majorly resistant to tetracycline (73.3%), cefazolin (66.6%), and trimethoprim (66.6%) (Table 2). However, the overall tetracycline resistance was 4.065 (95% CI, 1.477, 11.188) times more in DL. Unlikely, the odds of resistance against amoxycyclavulanate (OR, 0.437; 95% CI, 0.209, 0.912) and ampicillin (OR, 0.343; 95% CI, 0.163, 0.720) were still low in DL (Fig. 1).

Occurrence of KPC and MBL

The meropenem-resistant isolates from DL (13.3%; 10/75) and Cg (22.2%; 10/45) were subjected to KPC and MBL production tests. However, none of the isolates tested positive for KPC and/or MBL production.

MDR and MAR

MDR% varied significantly among the systems ($p < 0.05$), with DL harboring 6.439 (95% CI, 1.246, 33.283) times more MDR *E. coli* compared to Cg (Table 1). Further, majority of isolates from DL (97.7%; 88/90) and Cg (87.2%; 41/47) had MAR > 0.2, implying a high-risk source of contamination (Supplemental Data S1).

TABLE 1. SAMPLE-WISE OCCURRENCE OF *ESCHERICHIA COLI*, ESBL, AmpC, ESBL/AmpC, AND MDR *E. COLI*

System	Sample	Sample size	<i>E. coli</i> %	ESBL%	AmpC %	ESBL/AmpC %	MDR%
Deep litter	Egg	240	31.2 ^a	68 ^a	45.3	34.6 ^a	97.3 ^a
	Feed	12	16.6	50	50	50	100
	Water	12	50	100	83.3	83.3	100
	Dropping	12	41.6	60	0	0	100
	Swab	05	40	0	0	0	100
Cage	Egg	240	18.7 ^b	46.6 ^b	40	13.3 ^b	86.6 ^b
	Feed	12	—	—	—	—	—
	Water	12	—	—	—	—	—
	Dropping	12	16.6	100	100	100	100
	Swab	05	—	—	—	—	—

^{a,b}Different superscripts indicate a significant difference within the same sample type between two different systems of rearing ($p < 0.05$). ESBL, extended-spectrum beta-lactamase; MDR, multidrug resistant.

Hierarchical clustering of AmpC-producing *E. coli*

The dendrogram analysis of AmpC producers revealed the formation of seven major clusters, based on the predominance of resistance against the following antimicrobials; cluster 1, CTR, CTX, and TR; cluster 2, CTR, AMP, and CZ; cluster 3, CTX; cluster 4, CAZ; cluster 5, CTR, CAZ, CZ, and TR; cluster 6, TR; and cluster 7, AMC; respectively (Fig. 2).

Molecular characterization of antibiotic resistance genes in *E. coli* isolates

The study revealed a significantly higher presence of *bla*_{AmpC}, *qnrA/B/S*, *tetA/B*, and *aac(3)-IIa* and *aph(3)-Ia* genes in DL compared to Cg ($p < 0.05$) (Table 3). This was evident by higher odds of occurrence of *bla*_{AmpC} (5.046; 95% CI, 2.186, 11.65), *qnrA/B/S* (2.96; 95% CI, 1.313,

6.673), *tetA/B* (6.973; 95% CI, 1.997, 24.344), and *aac(3)-IIa* and *aph(3)-Ia* (2.776; 95% CI, 1.052, 7.326) in isolates from DL (Fig. 3). Furthermore, isolates carrying multiple antibiotic-resistant genes were also reported in the study (Supplemental Data S1).

Multilocus sequence typing of *E. coli* isolates

The present study reported ten distinct STs, viz., ST155, ST10, ST156, ST942, ST165, ST295, ST1286, ST2544, ST4162, and ST90 (Table 4). No clonal relationships were observed between the farms, and no new ST was reported. However, among the reported STs, ST10, 155, and 156 were found to be of public health importance, given their wide host spectrum and association with resistant gene determinants such as *mcr-1* and *bla*_{NDM} in previous studies.

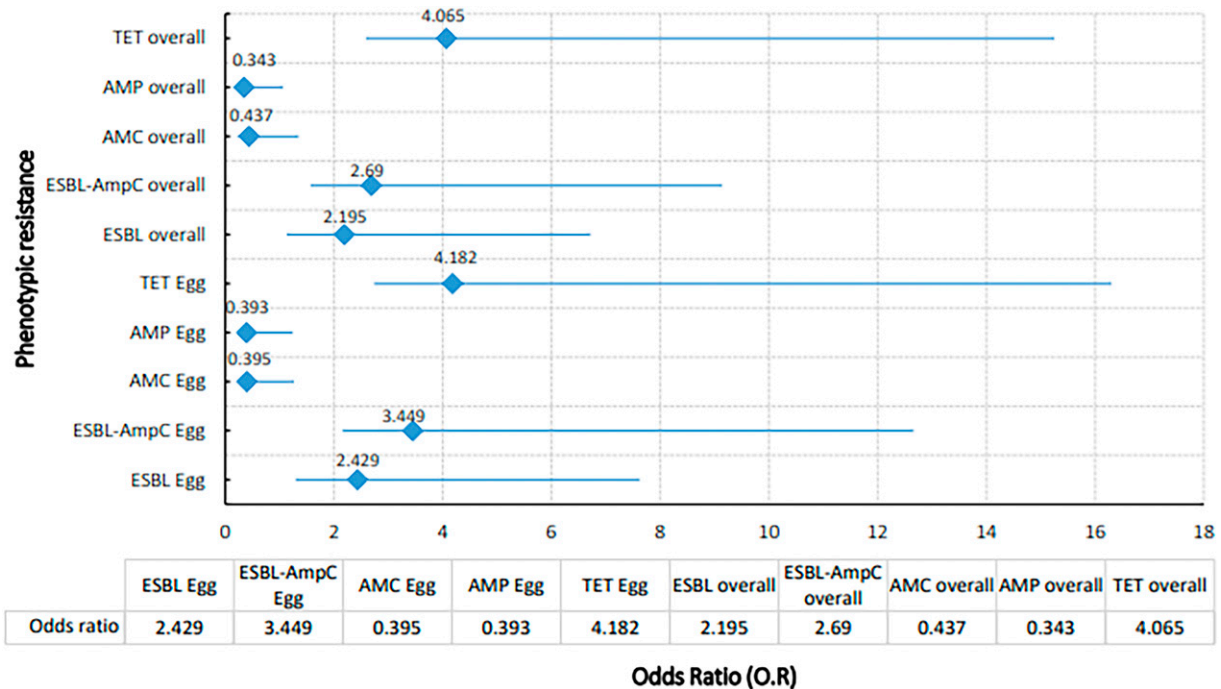


FIG. 1. Results from logistic regression modeling the risk of occurrence of phenotypic resistance (egg and overall) against poultry farming system, with cage system kept as the reference category. The X-axis represents the odds ratio on an exponential scale. The horizontal lines represent 95% confidence intervals (CI) for the odds ratio. AMC, amoxycyclavulanate; AMP, ampicillin; ESBL, extended-spectrum beta-lactamase-producing *Escherichia coli*; TET, tetracycline.

TABLE 2. PHENOTYPIC ANTIBIOGRAM PROFILE OF *ESCHERICHIA COLI* ISOLATES RECOVERED FROM DIFFERENT SYSTEMS OF REARING

System	Source	Isolates	% Resistance									
			Highest priority CIAs					High-priority CIAs				
			CTR	CAZ	LE	CIP	AK	GEN	TOB	AMC	AMP	MRP
Deep litter	Egg	75	56	52	54.6	53.6	48	14.6	46.6	24 ^a	44 ^a	13.3
	Feed	2	100	50	50	0	50	50	100	50	50	0
	Water	6	66.6	100	66.6	100	33.3	0	66.6	50	16.6	0
	Dropping	5	40	60	80	40	60	20	60	60	60	0
	Hand swab	2	50	0	50	0	0	100	0	0	0	0
Cage	Overall	90	55.5	67.7	53.3	54.4	46.6	16.6	48.8	27.7 ^c	42.2 ^c	11.1
	Egg	45	48.8	46.6	60	53.3	35.5	8.8	33.3	44.4 ^b	66.6 ^b	22.2
	Dropping	2	100	100	0	100	0	0	0	100	100	0
	Overall	47	51	48.9	57.4	55.3	34	8.5	31.9	46.8 ^d	68 ^d	21.2
			Highly important CIAs									
			TR	COT	TE	CZ	C					
			52	52.1	92 ^a	76	28					
			50	0	100	100	0					
			100	83.3	100	83.3	16.6					
			60	0	100	80	0					
			50	50	50	100	0					
			55.5	50	92.2 ^c	77.7	24.4					
			66.6	60	73.3 ^b	66.6	15.5					
			0	100	100	0	100					
			63.8	57.4	74.4 ^d	63.8	19.1					

^{a,b}Different superscripts indicate a significant difference in resistance % in the same sample type between two different rearing systems ($p < 0.05$).

^{c,d}Different superscripts indicate a significant difference in overall resistance % against the same antibiotic between two different rearing systems ($p < 0.05$).

AK, amikacin; AMC, amoxycyclavulanate; AMP, ampicillin; C, chloramphenicol; CIAs, critically important antimicrobials, antibiotics; CAZ, ceftazidime; COT, cotrimoxazole; CTR, ceftriaxone; CTX, cefotaxime; CIP, ciprofloxacin; CZ, ciprofloxacin; GEN, gentamicin; LE, levofloxacin; MRP, meropenem; TE, tetracycline; TOB, tobramycin; TR, trimethoprim.

Discussion

The contamination of farmed eggs varies depending on the poultry rearing conditions and management practices, particularly the BS measures implemented on farms. These measures are designed to prevent the introduction of external infections (external BS) and to limit the spread of pathogens within the farm (internal BS). Our study found significant differences between DL and caged (Cg) farms in terms of rearing practices. The majority of DL farms (58.3%) lacked on-farm BS measures (both external and internal). Additionally, the birds had unrestricted access to the outdoors, which led to eggs being laid near farm dust, dirt, feathers, and bird droppings—potential sources of bacterial contamination. Moreover, none of the farms cleaned either visibly clean or dirty eggs after laying, allowing any contamination on or within the eggs to persist from the farm to the consumer. In contrast, all Cg farms adhered to strict BS protocols (both external and internal) and housed birds in clean cages, with physical separation of eggs from the birds (i.e., cloaca, feathers, and droppings). Eggs were laid directly onto clean trays, minimizing contamination risks. These factors contributed to a higher presence of *E. coli* on eggs from DL farms compared to Cg systems. This observation aligns with findings from studies by De Reu et al. (2008) and Fahim et al. (2021), which reported higher bacterial loads on eggs from DL compared to Cg system.

Animal food products are subjected to risk of contamination with resistant bacteria and resistance genes at nearly all the stages of processing (Cabello, 2006; Dierikx et al., 2013) and pose a public health threat upon cross-contamination to clean surfaces, foods, and handlers (Cabello, 2006; Perreten et al., 2005). AMR in Gram-negative bacteria (GNB) like *E. coli* is largely driven by the production of ESBLs, AmpC enzymes, and carbapenemases. Beta-lactam resistance is a leading cause of treatment failure in resistant GNB infections (Laube et al., 2013; Rawat and Nair, 2010), necessitating the use of last resort antibiotics, viz., colistin, and carbapenems, thus promoting resistance to these drugs as well (Cantón et al., 2012). Our study observed that the occurrence of ESBL and ESBL/AmpC coproducers was 2.195 and 2.69 folds higher in eggs from DL. These findings corroborated with previous studies on poultry fecal and cloacal swab samples from India (Brower et al., 2017; Mondal et al., 2023; Senthilmiselvan et al., 2024); however, our reported occurrence was higher. Further, meropenem resistance (DL, 13.3%; Cg, 22.2%), and AmpC production (DL, 45.3%; Cg, 40%) were high in both systems, however, the variations were insignificant ($p > 0.05$). Our results indicate toward misuse of beta-lactams, especially third-generation cephalosporins in both systems, which could be higher in DL than Cg. This was supported by the fact that 58.3% of DL farms report self-prescribing antibiotics. Further free access of birds to outer surroundings in DL predisposes eggs to environmental ESBL and ESBL/AmpC coproducers as opposed to the Cg system.

Besides ESBLs, high resistance against tetracycline, fluoroquinolone, penicillin, sulfa drug, and aminoglycoside groups was evident in both systems; however, significant variation was noted only for tetracycline and penicillins. The resistance levels were lower compared to hatchery eggs, which showed 100% tetracycline and 93.47% ampicillin resistance, as reported by Sharan et al. (2023). We sought to establish a link between the reported on-farm antimicrobial usage (AMU)

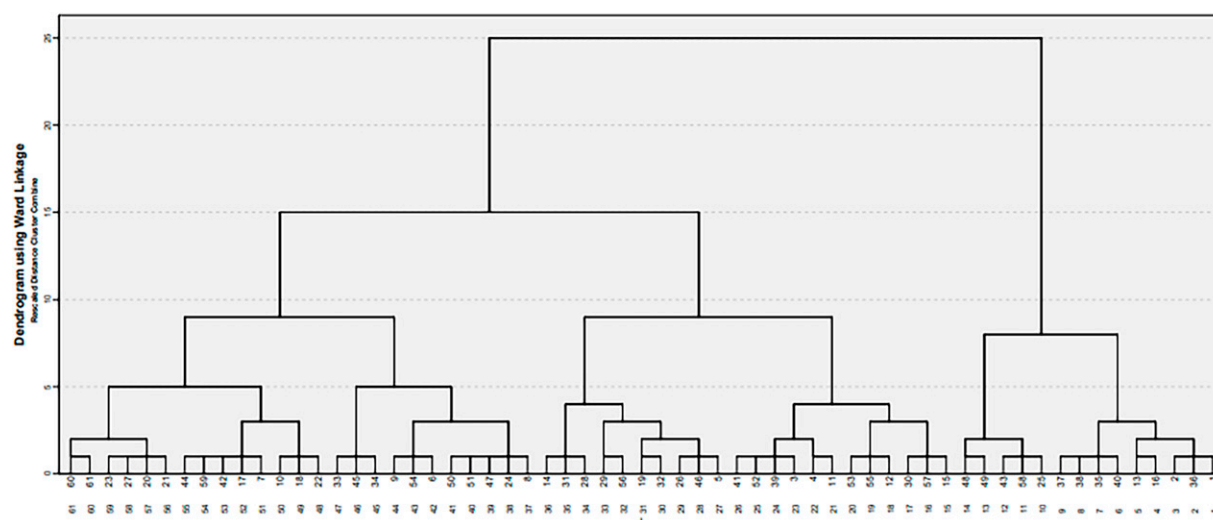


FIG. 2. Hierarchical clustering of AmpC-producing *Escherichia coli* based on phenotypic resistance against CTR, CAZ, CTX, CZ, AMP, AMC, and TR, using Ward's method and squared Euclidean distance measure. AMC, amoxycyclavulanate; AMP, ampicillin; CAZ, ceftazidime; CTR, ceftriaxone; CTX, cefotaxime; CZ, cefazolin; TR, trimethoprim.

and the resistance profiles across farms. DL farms reported using cephalaxin (83.3%), neomycin (75%), fluoroquinolones (66.6%), tetracycline (66.6%), and sulfonamides (66.6%). In contrast, Cg farms used cephalaxin (16.6%) and fluoroquinolones (41.6%). However, the reported AMU does not fully explain the differences in resistance profiles between the two systems. Notably, none of the farms reported using third-gen. cephalosporins, carbapenems, or amphenicols, yet resistance to these drug classes was still high.

Additionally, the MDR status of both systems was fairly high: 97.7%, DL; 87.2%, Cg, with 100% MDR from poultry droppings in both systems. Since all farms used droppings as a soil amendment on agricultural farms, the risk of transmission of these MDR *E. coli* to the environment was high. Additionally, the human handlers and consumers were at an escalated risk of counteracting ESBL, ESBL/AmpC, and MDR *E. coli* following improper handling, processing, and consumption of contaminated eggs. Interestingly, despite high overall CAZ resistance (67.7%, DL; 48.9%, Cg), the reported occurrence of CTX-M-15 gene was observed to be

low (2.6%, DL; 2.2%, Cg). This could be due to the presence of another CTX-M group 1 variant, CTX-M 27, which is also responsible for CAZ hydrolysis (Castanheira et al., 2021).

The multilocus sequence typing (MLST) analysis of MDR isolates from both systems revealed genetic diversity, supported by distinct STs, with no clonal relationships. However, STs from poultry droppings in DL, and Cg farms shared the same clonal complex, that is, ST155. ST155 was previously reported from live bird markets, farm environments, and handlers in Nigeria (Aworh et al., 2019); as a clonal ST in cattle and pigs with MBL determinants (Tewari et al., 2019) in India; food animals across Europe (Ewers et al., 2011); and humans (Matamoros et al., 2017) worldwide. Furthermore, ST10 and ST155 were associated with colistin-resistant *E. coli* in chicken fecal samples from Lebanon (Al-Mir et al., 2021). Further, we found ST156 from Cg egg, which has been associated with *bla*_{NDM-1} and *bla*_{CTX-M-15} *E. coli* in humans and poultry (Giufre et al., 2012; Mush-taq, et al., 2011), *bla*_{NDM-5} in layer feces (Tang et al., 2019)

TABLE 3. MOLECULAR CHARACTERIZATION OF *ESCHERICHIA COLI* ISOLATES FROM DIFFERENT SYSTEMS OF POULTRY REARING

		% Resistance genes								
System	Sample	Isolates	<i>bla</i> _{TEM}	<i>bla</i> _{SHV}	<i>bla</i> _{CTXM-15}	<i>bla</i> _{AmpC}	<i>qnr</i> (A/B/S)	<i>tet</i> (A/B)	<i>sul</i> (1&2)	<i>aac</i> (3)- <i>Ila</i> <i>aph</i> (3)- <i>Ia</i>
Deep litter	Egg	75	26.6	2.6	2.6	56 ^a	42.3 ^a	30.6 ^a	13.3	24
	Feed	2	0	0	0	50	0	100	0	50
	Water	6	50	0	0	50	66.6	50	16.6	50
	Dropping	5	60	0	0	60	80	60	40	60
	Hand swab	2	0	0	0	0	50	0	0	50
	Overall	90	28.8	2.2	2.2	54.4 ^a	46.6 ^a	34.4 ^a	14.4	28.8
Cage	Egg	45	42.25	8.8	2.2	17.7% ^b	2.2 ^b	8.8 ^b	2.2	11.1
	Dropping	2	0	0	0	100	100	0	0	50
	Overall	47	40.40	8.5	2.1	21.2 ^b	6.38 ^b	8.5 ^b	2.1	12.7

^{a,b}Different superscript in the same sample source and AMR genes between different systems of rearing indicate a significant difference ($p < 0.05$).

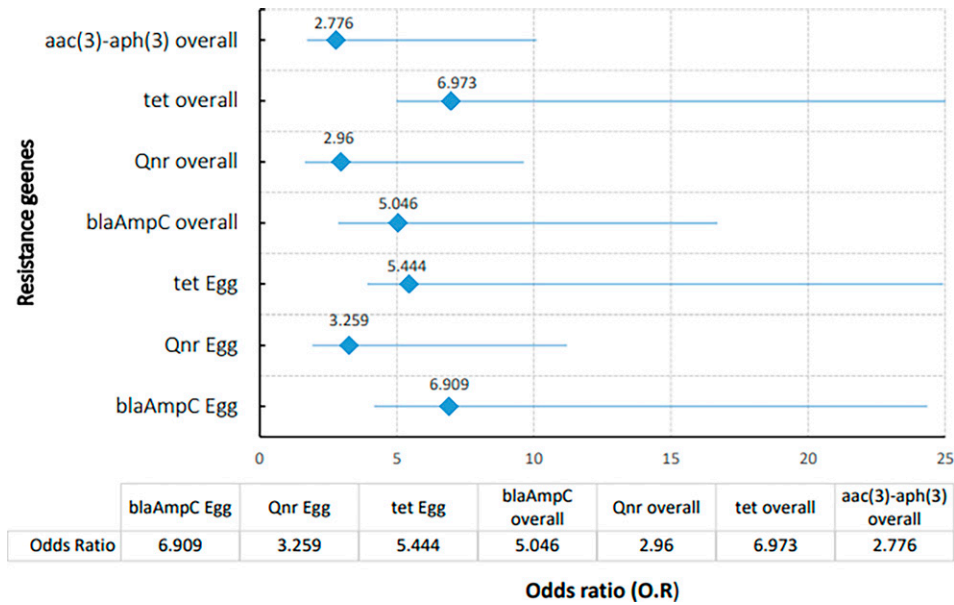


FIG. 3. Results from logistic regression modeling the risk of occurrence of resistance genes (egg and overall) versus poultry farming system, with cage system kept as the reference category. The X-axis represents the odds ratio on an exponential scale. The horizontal lines represent 95% confidence intervals (CI) for the odds ratio.

and *mcr-1* and *bla*_{NDM-5} in Muscovy duck (Yang et al., 2016). The presence of such STs from egg and farm environment in our study emphasizes on need of implementation of farm BS, continuous surveillance of eggs, and advisory over safe handling/processing of eggs to break the chain of transmission.

Conclusion

The present study reported high MDR, ESBL, and ESBL/AmpC-producing *E. coli* from DL system, in addition to high tetracycline resistance. Additionally, DL system was predominated by *bla*_{AmpC}, *tetA/B*, *qnrA/B/S*, and *aac(3)-IIa -aph(3)-Ia* genes. Further AmpC *E. coli* clustered together based on resistance against cephalosporins, penicillins, and sulfa drugs. The MLST profile of MDR *E. coli* isolates marked the presence of STs 10, 155, and 156, which were associated with colistin and metallo-beta-lactamase determinants in previous studies. Therefore, the present study addresses the potential of eggs and farm

environment as vehicles of MDR to human handlers upon cross-contamination at farm and household, and therefore highlights the need for implementation of strict BS and hygienic measures while production, collection and handling of eggs.

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Authors' Contributions

Conceptualization: S.T., R.S., J.S.B., and A.K.A. Methodology: S.T., R.S., J.S.B., A.K.A., R.S.A., and J.S. Experimentation: S.T. Writing—original draft preparation: S.T. Writing—reviewing and editing: R.S. and J.S.B. Sampling: S.T. and J.S. Statistical analysis: S.T. Supervision: R.S. and A.K.A. Project administration: A.K.A.

TABLE 4. MULTILOCUS SEQUENCE TYPING (MLST) PROFILE OF MDR *ESCHERICHIA COLI* ISOLATES

Accession No.	Sample	Phenotypic resistance	ST	Clonal complex
CP123272	DL Egg	CTR, LE, TE, TR, COT, AK, TOB	ST10	ST10
CP125369	DL Egg	ESBL, TE, CZ, TR, COT	ST6836	ST165
CP125621	DL Egg	ESBL, CTR, LE, CIP, AMC, AK, GEN	ST1286	—
CP1232773	DL Egg	ESBL, AmpC, CTR, AMP, AMC, TE, CZ, AK, TOB	ST942	—
CP125622	DL Dropping	ESBL, AmpC, LE, TE, CZ, TR, AK, GEN, TOB	ST4162	ST155
CP123274	Cg Egg	CTR, CZ, AK	ST295	—
CP123276	Cg Egg	LE, CIP, AMP, AMC, TE, CZ, TR, COT	ST90	ST23
CP123277	Cg Egg	ESBL, LE, CIP, AMP, TE, TR, COT,	ST2544	—
CP123271	Cg Egg	ESBL, AmpC, CTR, CIP, LE, AMP, TE, CZ, TR, COT, AK, TOB, MRP	ST156	ST156
CP123275	Cg Dropping	ESBL, AmpC, CIP, AMP, AMC, TE, TR, COT, C	ST155	ST155

AK, amikacin; AMC, amoxycyclavulanate; AMP, ampicillin; C, chloramphenicol; COT, co-trimoxazole; CTR, ceftriaxone; CIP, ciprofloxacin; CZ, cefazolin; GEN, gentamicin; LE, levofloxacin; MRP, meropenem; TE, tetracycline; TOB, tobramycin; TR, trimethoprim.

Disclosure Statement

The authors declare no conflict of interest.

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Supplementary Material

Supplemental Data S1

References

- Al-Mir H, Osman M, Drapeau A, et al. WGS analysis of clonal and plasmidic epidemiology of colistin-resistance mediated by *mcr* genes in the poultry sector in Lebanon. *Front Microbiol* 2021;12:624194; doi: 10.3389/fmicb.2021.624194
- Aworh MK, Kwaga J, Okolocha E, et al. Prevalence and risk factors for multi-drug resistant *Escherichia coli* among poultry workers in the Federal Capital Territory, Abuja, Nigeria. *PLoS ONE* 2019; 14(11):e0225379; doi: 10.1371/journal.pone.0225379
- Brower CH, Mandal S, Hayer S, et al. The prevalence of extended-spectrum beta-lactamase-producing multidrug-resistant *Escherichia coli* in poultry chickens and variation according to farming practices in Punjab, India. *Environ Health Perspect* 2017; 125(7):e077015; doi: 10.1289/EHP292
- Cabello FC. Heavy use of prophylactic antibiotics in aquaculture: A growing problem for human and animal health and for the environment. *Environ Microbiol* 2006;8(7):1137–1144; doi: 10.1111/j.14622920.2006.01054.x
- Cantón R, González-Alba JM, Galán JC. CTX-M enzymes: Origin and diffusion. *Front Microbiol* 2012;3:110; doi: 10.3389/fmicb.2012.00110
- Castanheira M, Simmer PJ, Bradford PA. Extended-spectrum β -lactamases: An update on their characteristics, epidemiology and detection. *JAC Antimicrob Resist* 2021;3(3):dlab092; doi: 10.1093/jacamr/dlab092
- Clinical and Laboratory Standards Institute (CLSI). Antimicrobial Susceptibility Testing Standards M100 (Performance Standards for Antimicrobial Susceptibility Testing), 30th Edition. CLSI; 2020. <https://clsi.org/standards/products/microbiology/documents/m100/CLSI>
- Crowther-Gibson P, Govender N, Lewis DA, et al. Human infections and antibiotic resistance. *South African Medical Journal* 2011;101(sup-1):567–578. <https://hdl.handle.net/10520/EJC67697>
- De Reu K, Messens W, Heyndrickx M, et al. Bacterial contamination of table eggs and the influence of housing systems. *World's Poultry Science Journal* 2008;64(1):5–19; doi: 10.1017/S00439333907001687
- Department of Biotechnology. Indian priority Pathogen list, 2021. Available from: https://dbtindia.gov.in/sites/default/files/IPPL_final.pdf [Last accessed: June 3, 2024].
- Dierikx C, van der Goot J, Fabri T, et al. Extended-spectrum β -lactamase-and AmpC- β -lactamase-producing *Escherichia coli* in Dutch broilers and broiler farmers. *J Antimicrob Chemother* 2013;68(1):60–67; doi: 10.1093/jac/dks349
- Enterobase. Enterobase. Available from: http://enterobase.warwick.ac.uk/species/ecoli/search_strains last [Last accessed: June 3, 2024].
- Ewers C, Grobbel M, Bethe A, et al. Extended-spectrum beta-lactamases-producing gram-negative bacteria in companion animals: Action is clearly warranted!. *Berl Munch Tierarztl Wochenschr* 2011;124(3–4):94–101; doi: 10.2376/0005-9366-124-94
- Fahim KM, Khalf MAM, Nader SM, et al. Impacts of housing and storage environments on physical quality and the potential public health risks of chicken table eggs. *Advances in Animal and Veterinary Science* 2021;9(8):1176–1189; doi: 10.17582/journal.aavs/2021/9.8.1176.1189
- Food and Agriculture Organization Corporate Statistical Database [FAOSTAT]. FAOSTAT. 2021. Available from: <http://www.fao.org/faostat/en/#data/QC>
- Founou RC, Blocker AJ, Noubom M, et al. The COVID-19 pandemic: A threat to antimicrobial resistance containment. *Future Sci OA* 2021;7(8):FSO736; doi: 10.2144/fsoa-2021-0012
- Giufre M, Graziani C, Accogli M, et al. *Escherichia coli* Study Group. *Escherichia coli* of human and avian origin: Detection of clonal groups associated with fluoroquinolone and multi-drug resistance in Italy. *J Antimicrob Chemother* 2012;67(4): 860–867; doi: 10.1093/jac/dkr565
- Krumperman PH. Multiple antibiotic resistance indexing of *Escherichia coli* to identify high-risk sources of fecal contamination of foods. *Appl Environ Microbiol* 1983;46(1):165–170; doi: 10.1128/aem.46.1.165-170.1983
- Laube H, Friese A, Von Salviati C, et al. Longitudinal monitoring of extended-spectrum-beta-lactamase/AmpC-producing *Escherichia coli* at German broiler chicken fattening farms. *Appl Environ Microbiol* 2013;79(16):4815–4820; doi: 10.1128/AEM.00856-13
- Llor C, Bjerrum L. Antimicrobial resistance: Risk associated with antibiotic overuse and initiatives to reduce the problem. *Ther Adv Drug Saf* 2014;5(6):229–241; doi: 10.1177/2042098614554919
- Matamoros S, Van Hattem JM, Arcilla MS, et al. Global phylogenetic analysis of *Escherichia coli* and plasmids carrying the *mcr-1* gene indicates bacterial diversity but plasmid restriction. *Sci Rep* 2017;7(1):15364; doi: 10.1038/s41598-017-15539-7
- Mondal T, Dey S, Samanta I, et al. Characterization of extended-spectrum and pAmpC Beta-lactamases producing *Escherichia coli* isolated from Chicken Meat in West Bengal. *IJVP* 2023; 9(2):27–32; doi: 10.62418/ijvp.9.2.2023.27-32
- Mushtaq S, Irfan S, Sarma JB, et al. Phylogenetic diversity of *Escherichia coli* strains producing NDM-type carbapenemases. *J Antimicrob Chemother* 2011;66(9):2002–2005; doi: 10.1093/jac/dkr226
- Perreten V, Vorlet-Fawer L, Slickers P, et al. Microarray-based detection of 90 antibiotic resistance genes of gram-positive bacteria. *J Clin Microbiol* 2005;43(5):2291–2302; doi: 10.1128/jcm.43.5.2291-2302.2005
- Poulou A, Grivakou E, Vrioni G, et al. Modified CLSI extended-spectrum β -lactamase (ESBL) confirmatory test for phenotypic detection of ESBLs among Enterobacteriaceae producing various β -lactamases. *J Clin Microbiol* 2014;52(5):1483–1489; doi: 10.1128/jcm.03361-13
- Rawat D, Nair D. Extended-spectrum β -lactamases in Gram Negative Bacteria. *J Glob Infect Dis* 2010;2(3):263–274; doi: 10.4103/0974-777X.68531
- Sakthikarthikeyan S, Sivakumar M, Manikandan R, et al. Prevalence and molecular characterization of multidrug-resistant

- ESBL-producing *E. coli* in commercial poultry. IJAR 2024; 1(Of); doi: 10.18805/IJAR.B-5166
- Senthamilselvan B, Rameshkumar MR, Syed Madani Z, et al. Molecular analysis of ESBL-and AmpC-producing Enterobacteriaceae in fecal samples from broiler and backyard chickens. J King Saud University-Science 2024;36(6):103191; doi: 10.1016/j.jksus.2024.103191
- Sharan M, Dhaka P, Bedi JS, et al. Characterization of chicken eggs associated *Escherichia coli* and *Staphylococcus aureus* for biofilm production and antimicrobial resistance traits. Anim Biotechnol 2023;34(8):3533–3544; doi: 10.1080/10495398.2023.2171423
- Sharma I, Bhat MA, Taku A, et al. Determination of prevalence and molecular characterization of extended spectrum beta-lactamase (ESBL) producing *Escherichia coli* in poultry from J & K, India. J Entomol Zool Stud 2021;9: 2108–2211.
- Tang B, Chang J, Cao L, et al. Characterization of an NDM-5 carbapenemase-producing *Escherichia coli* ST156 isolate from a poultry farm in Zhejiang, China. BMC Microbiol 2019;19(1):1–12; doi: 10.1186/s12866-019-1454-2
- Tewari R, Mitra S, Ganaie F, et al. Dissemination and characterisation of *Escherichia coli* producing extended-spectrum β -lactamases, AmpC β -lactamases and metallo- β -lactamases from livestock and poultry in Northeast India: A molecular surveillance approach. J Glob Antimicrob Resist 2019;17:209–215; doi: 10.1016/j.jgar.2018.12.025
- World Health Organization [WHO]. WHO Bacterial Priority Pathogens List, 2024: Bacterial Pathogens of Public Health Importance to Guide Research, Development and Strategies to Prevent and Control Antimicrobial Resistance. Geneva: World Health Organization; 2024.
- Wu C, Wang Y, Shi X, et al. Rapid rise of the ESBL and mcr-1 genes in *Escherichia coli* of chicken origin in China, 2008–2014. Emerg Microbes Infect 2018;7(1):30; doi: 10.1038/s41426-018-0033-1
- Yang RS, Feng Y, Lv XY, et al. Emergence of NDM-5- and MCR-1-producing *Escherichia coli* clones ST648 and ST156 from a single muscovy duck (*Cairina moschata*). Antimicrob Agents Chemother 2016;60(11):6899–6902; doi: 10.1128/aac.01365-16

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